Chapter 1: Essential Ideas of Chemistry

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PHY201

Lecture 1

Part I

The Scope of Chemistry

Outline of Part I: The Scope of Chemistry

Outline of Part I: The Scope of Chemistry

What is Chemistry?

Definition: Chemistry

Chemistry is the study of the composition, properties, and interactions of matter.

Why study it?

- Coffee brewing, digestion, soap lathering, fuel burning — all chemistry.
- Every manufactured object you own relies on chemical knowledge.
- Chemistry is the *central science*: it bridges physics, biology, engineering, medicine, and environmental science.

- A typical smartphone contains $\sim$30% of all naturally occurring elements.
- Circuit boards, batteries, screens, and cases are all the result of chemistry.

"Chemistry"

chemistry [ˈkɛm.ɪ.stɹi] *noun* – English

The branch of natural science that deals with the composition and constitution of substances and the changes that they undergo as a consequence of alterations in the constitution of their molecules.

↑

χύμα [kʰý.ma] *noun* – Ancient Greek

that which is poured out or flows, fluid; ingot, bar

The Three Domains of Chemistry

Macroscopic

Observable, everyday-scale phenomena.

Examples: ice melting, food digesting, gasoline burning

Microscopic

Molecular and atomic structures too small to see directly.

Examples: molecular bonds, electron behavior

Symbolic

Chemical symbols, formulas, and equations representing matter.

Examples: H_2O, CO_2, balanced equations

A key skill in chemistry is moving fluidly between all three domains:

- "Water boils" (macroscopic)
- H_2O molecules escape (microscopic)
- $H_2O(\ell) \rightarrow H_2O(g)$ (symbolic)

The Scientific Method

1. **Observe** the natural world and ask a question.
2. **Form a hypothesis**: a tentative, testable explanation.
3. **Design and conduct experiments** to test the hypothesis.
4. **Analyze data** and draw conclusions.
5. **Refine** the hypothesis; if broadly confirmed over many experiments, may advance to a theory or law.

The scientific method is a **cycle**, not a straight line. Unexpected results drive new hypotheses.

Hypothesis, Theory, and Law

Hypothesis

A **hypothesis** is a tentative explanation of observations; it guides further experimentation. *e.g. "The patient's fever is caused by a bacterial infection."*

Scientific Theory

A **theory** is a well-substantiated, comprehensive, testable explanation of a class of natural phenomena, supported by extensive evidence. *e.g. Atomic theory, kinetic molecular theory*

Scientific Law

A **law** summarizes vast experimental observations and describes or predicts natural phenomena — but does not explain *why*. *e.g. Law of conservation of matter*

☠️ In science, a theory never "becomes" a law. They describe different things.

Check Your Understanding

Classify each statement as a **hypothesis**, a **scientific theory**, or a **law**:

(a) Solids expand when heated.

(b) Matter is made of particles too small to see.

(c) Drinking less sugar will reduce my risk of cavities.

Come to lecture to see the answer.

Outline of Part I: The Scope of Chemistry

What is Matter?

Definition: Matter

Matter is anything that occupies space and has **mass**.

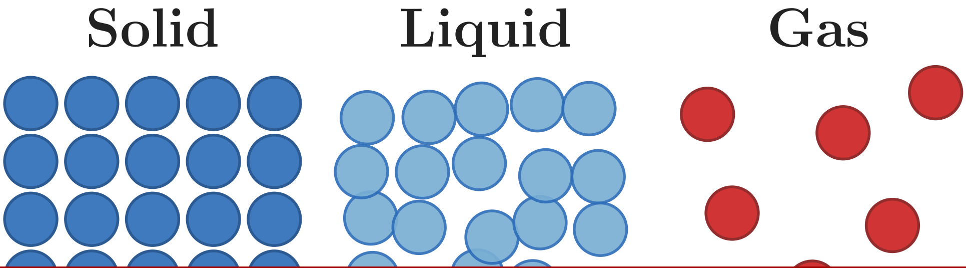
Mass vs. Weight

- **Mass**: Amount of matter in an object. Does **not** change with location.
- **Weight**: Force that gravity exerts on an object. Changes with gravitational field.

An astronaut's **mass** is the same on Earth and the Moon.

Her **weight** on the Moon is $\frac{1}{6}$ of her Earth weight ($\because g_{\text{Moon}} \approx \frac{1}{6}\, g_{\text{Earth}}$).

The States of Matter



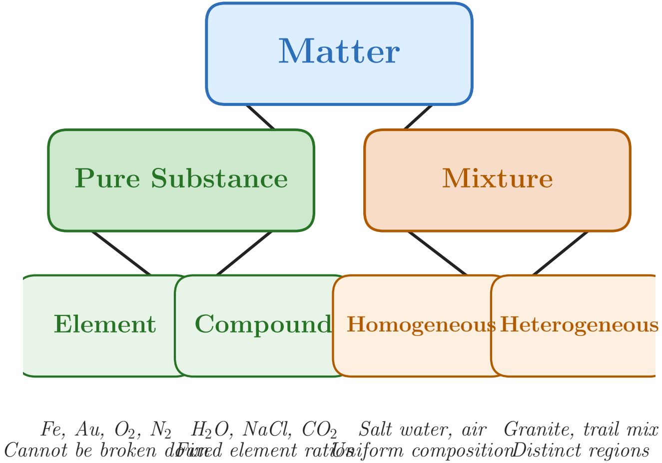
Check Your Understanding

Which property changes when you move from Earth to the Moon?

(a) Your mass (b) Your weight

Come to lecture to see the answer.

Classification of Matter



Check Your Understanding

Classify each as an **element**, **compound**, **homogeneous mixture**, or **heterogeneous mixture**:

(a) Air in this classroom

(b) Pure liquid water

(c) 18-karat gold (gold + copper, uniformly blended)

(d) A bowl of trail mix

Come to lecture to see the answer.

Atoms and Molecules

- An **atom** is the smallest particle of an element that retains the chemical properties of that element.
- A **molecule** is two or more atoms joined by chemical bonds.
- Some elements naturally exist as **diatomic molecules** — pairs of identical atoms bonded together.

The Seven Diatomic Elements — (Memorize these)

Hydrogen	H_2	Fluorine	F_2	Iodine	I_2
Nitrogen	N_2	Chlorine	Cl_2		
Oxygen	O_2	Bromine	Br_2		

Memory aid: **"Have No Fear Of Ice Cold Beer"** $\Rightarrow$ H, N, F, O, I, Cl, Br

Check Your Understanding

You have a sample of pure water (H_2O).

Is water an **element** or a **compound**?

Can it be broken down into simpler substances?

Come to lecture to see the answer.

Law of Conservation of Matter

Law of Conservation of Matter

There is no detectable change in the **total quantity of matter** when matter converts from one type to another (chemical change) or changes between states (physical change).

Example: Iron Nail Rusting

An iron nail has mass 23.2 g. After rusting, its mass is 24.1 g.

Where does the extra mass come from?

The nail gains mass by chemically combining with **oxygen from the air** to form iron oxide (rust).
Mass of nail + mass of oxygen consumed = mass of rust.   Matter is conserved.

Part II

Properties of Matter

Outline of Part II: Properties of Matter

Outline of Part II: Properties of Matter

Physical and Chemical Properties

Physical Properties

Definition: Physical Property

A **physical property** is a characteristic of matter observable or measurable *without* changing the chemical identity of the substance.

Examples of Physical Properties

- Color, odor, luster, hardness
- Density, melting point, boiling point
- Electrical conductivity, solubility, state of matter

Physical properties describe the substance itself; no chemical reaction is needed.

Chemical Properties

Definition: Chemical Property

A **chemical property** describes a substance's ability (or inability) to transform into a *different* substance through chemical change.

Examples of Chemical Properties

- Flammability (wood burns; gold does not)
- Toxicity (carbon monoxide is poisonous)
- Acidity/basicity (vinegar reacts with baking soda)
- Reactivity with water, oxygen, or acids

Chemical properties are only observed **during a chemical change**, when the substance interacts with something else and is transformed.

Physical vs. Chemical Changes

Physical Change

A change in state or appearance that does **not alter the chemical identity**.

Examples:

- Ice melting to liquid water
- Sugar dissolving in coffee
- Grinding chalk into powder
- Magnetizing a metal rod

Chemical Change

Produces one or more substances with **different chemical identities**.

Examples:

- Iron rusting ($Fe + O_2 \rightarrow Fe_2O_3$)
- Wood burning (combustion)
- Cooking an egg
- Milk souring

Signs of a chemical change: production of heat or light, gas formation, precipitate formation, or a permanent color change.

Check Your Understanding

Classify each as a **physical** or **chemical** change:

(a) Slicing a loaf of bread

(b) Burning gasoline in a car engine

(c) Water evaporating from a puddle

(d) A silver spoon tarnishing over time

Come to lecture to see the answer.

Extensive vs. Intensive Properties

Extensive Properties

Depend on the **quantity of matter** present.

Examples: mass, volume, length, amount of heat

Intensive Properties

Do **not** depend on the amount of matter.

Examples: temperature, density, color, melting point, boiling point

A **gallon** of water has **greater mass** than a **cup** of water.

Both a gallon of water and a cup of water have the same density and boiling point.

Intensive properties can be used to **identify** substances, regardless of sample size.
e.g. density of gold (Au) is **always** $19.3\ \frac{\text{g}}{\text{cm}^3}$.

Check Your Understanding

Is temperature an **extensive** or **intensive** property?

Is the amount of heat stored in a pot of boiling water **extensive** or **intensive**?

Come to lecture to see the answer.

Chemical Reactions — Energy Transfer

Definition: Energy

Energy is the capacity to do work or transfer heat.

Kinetic Energy

Energy of **motion**.

Examples:

- A moving car
- Flowing water
- Vibrating atoms in a hot object

Potential Energy

Energy of **position** or stored energy.

Examples:

- A ball held at height
- A compressed spring
- Chemical bonds in fuel

Chemical reactions involve the conversion of **potential energy stored in bonds** into kinetic energy (or vice versa).
e.g. burning gasoline releases stored chemical potential energy as heat and motion.

Part III

Measurement

Outline of Part III: Measurement

Outline of Part III: Measurement

Measurements

Measurement Uncertainty, Accuracy, and Precision

Why Measure?

A measurement provides three types of information:

1. A **magnitude** (a number)
2. A **unit** (a standard for comparison)
3. An indication of **uncertainty** (precision of the instrument)

A number without units is meaningless — or dangerous.

"Give the patient 100" — 100 mg (effective) vs. 100 g (lethal).

SI Base Units

The **International System of Units (SI)** is the globally accepted measurement standard, based entirely on powers of 10.

Property	Unit Name	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Amount of substance	mole	mol

Common Metric Prefixes

Prefix	Symbol	Value	Example (typical)
tera	T	10^{12}	terawatt (TW) $\sim 10^{12}$ W — powerful laser output
giga	G	10^{9}	gigahertz (GHz) $\sim 10^{9}$ Hz — microwave frequency
mega	M	10^{6}	megacurie (MCi) $\sim 10^{6}$ Ci — high radioactivity
kilo	k	10^{3}	kilometer (km) $= 10^{3}$ m — about 0.6 mile
—	—	10^{0}	(unity)
deci	d	10^{-1}	deciliter (dL) $= 10^{-1}$ L — less than half a soda
centi	c	10^{-2}	centimeter (cm) $= 10^{-2}$ m — fingertip thickness
milli	m	10^{-3}	millimeter (mm) $= 10^{-3}$ m — flea height at shoulders
micro	µ	10^{-6}	micrometer (µm) $= 10^{-6}$ m — microscope detail
nano	n	10^{-9}	nanogram (ng) $= 10^{-9}$ g — tiny dust speck
pico	p	10^{-12}	picofarad (pF) $= 10^{-12}$ F — small radio capacitor
femto	f	10^{-15}	femtometer (fm) $= 10^{-15}$ m — proton size scale

Check Your Understanding

Without calculating, which is **larger**?

(a) 1 km or 1 mm?

(b) 1 Mg or 1 kg?

(c) 1 µs or 1 ns?

Come to lecture to see the answer.

Density

Definition: Density

Density is an *intensive* property equal to the mass of a substance divided by the volume it occupies.

$$d = \frac{m}{V}$$

Common units:
$\frac{g}{mL}$ or $\frac{g}{cm^3}$ (for solids & liquids)
$\frac{g}{L}$ (for gases)

Substance	**Density**
Air (25 °C)	$1.20\ \frac{g}{L}$
Gasoline	$740\ \frac{g}{L} = 0.74\ \frac{g}{mL}$
Water (25 °C)	$1000\ \frac{g}{L} = 1.00\ \frac{g}{mL}$
Mercury	$1360\ \frac{g}{L} = 1.36\ \frac{g}{mL}$
Gold	$1930\ \frac{g}{L} = 1.93\ \frac{g}{mL}$

Example: Is It Gold?

Density Calculation — (chem2e_ch1_ex1.4)

You suspect a cube is not gold. Its edge length is 2.00 cm and its mass is 90.7 g. What is its density? Is it gold? (Gold: $19.3\,\frac{\mathrm{g}}{\mathrm{cm}^3}$)

Given:

$$\ell = 2.00\,\mathrm{cm}$$
$$m = 90.7\,\mathrm{g}$$

$$V = \ell^3 = (2.00\,\mathrm{cm})^3 = 8.00\,\mathrm{cm}^3$$

$$d = \frac{m}{V} = \frac{90.7\,\mathrm{g}}{8.00\,\mathrm{cm}^3}$$

$$\boxed{d = 11.3\,\frac{\mathrm{g}}{\mathrm{cm}^3}}$$

Gold's density is $19.3\,\frac{\mathrm{g}}{\mathrm{cm}^3}$. This cube is much less dense — consistent with **lead** ($11.3\,\frac{\mathrm{g}}{\mathrm{cm}^3}$). Not gold!

Unit Conversions: The Conversion Factor Method

Convert a quantity by multiplying by a **conversion factor** (a fraction equal to 1) so that unwanted units cancel.

Example: How many km are in $72\,\mathrm{cm}$?

$$72\,\mathrm{cm}\left(\frac{1\,\mathrm{m}}{100\,\mathrm{cm}}\right)\left(\frac{1\,\mathrm{km}}{1000\,\mathrm{m}}\right) = \boxed{7.2\times10^{-4}\,\mathrm{km}}$$

Example: How many g are in $3\,\mathrm{Mg}$? (Note: $3\,\mathrm{Mg}\neq 3\,\mathrm{mg}$)

$$3\,\mathrm{Mg}\left(\frac{10^6\,\mathrm{g}}{1\,\mathrm{Mg}}\right) = \boxed{3\times10^{6}\,\mathrm{g}}$$

Unit Conversions: Multi-Step

Example: How many mL are in $1\,\mathrm{m}^3$?

$$1\,\mathrm{m}^3\left(\frac{100\,\mathrm{cm}}{1\,\mathrm{m}}\right)^3\left(\frac{1\,\mathrm{mL}}{1\,\mathrm{cm}^3}\right)=\boxed{1\times10^6\,\mathrm{mL}}$$

Example: How many ks are in one year?

$$1\,\mathrm{yr}\left(\frac{365\,\mathrm{d}}{1\,\mathrm{yr}}\right)\left(\frac{24\,\mathrm{h}}{1\,\mathrm{d}}\right)\left(\frac{60\,\mathrm{min}}{1\,\mathrm{h}}\right)\left(\frac{60\,\mathrm{s}}{1\,\mathrm{min}}\right)\left(\frac{1\,\mathrm{ks}}{1000\,\mathrm{s}}\right)\approx\boxed{31.5\,\mathrm{ks}}$$

Set up the chain so that unwanted units appear once in the numerator and once in the denominator so that they cancel.

Check Your Understanding

A soft drink can contains 355 mL of liquid.

Set up the conversion to express this volume in **liters**. What is the answer?

Come to lecture to see the answer.

Temperature Scales

	°F	**°C**	**K**
Water boils	212	100	373.15
Body temp	98.6	37.0	310.15
Water freezes	32	0	273.15
Absolute zero	−459.67	−273.15	0

Conversion Formulas

$$T(^\circ\mathrm{F}) = \tfrac{9}{5}\,T(^\circ\mathrm{C}) + 32$$

$$T(^\circ\mathrm{C}) = \tfrac{5}{9}\left[T(^\circ\mathrm{F}) - 32\right]$$

$$T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273.15$$

Kelvin uses **no degree symbol**.
A change of $1\,\mathrm{K}$ = a change of $1\,^\circ\mathrm{C}$.

Example: Temperature Conversions

Body Temperature and Pizza Oven — (chem2e_ch1_ex1.11–1.12)

(a) Normal body temperature is 37.0 °C. Convert to °F and K.
(b) A pizza oven is set to 450 °F. Convert to °C.

(a) Body temp:

$$T(^\circ\mathrm{F}) = \frac{9}{5}(37.0) + 32$$

$$\boxed{T(^\circ\mathrm{F}) = 98.6\,^\circ\mathrm{F}}$$

$$T(\mathrm{K}) = 37.0 + 273.15$$

$$\boxed{T(\mathrm{K}) = 310.2\,\mathrm{K}}$$

(b) Pizza oven:

$$T(^\circ\mathrm{C}) = \frac{5}{9}(450 - 32)$$

$$= \frac{5}{9}(418)$$

$$\boxed{T(^\circ\mathrm{C}) \approx 232\,^\circ\mathrm{C}}$$

Check Your Understanding

Liquid nitrogen has a boiling point of 77 K.

What is this temperature in °C? Is it hotter or colder than the freezing point of water?

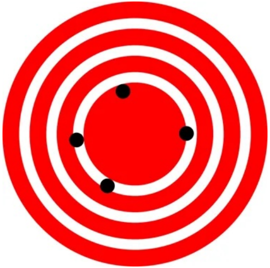
Come to lecture to see the answer.

Outline of Part III: Measurement

Importance of Measurements

Science is based on observation and experiment - that is, on **measurements**.

Accuracy of a Measurement



Accuracy is how close a measurement is to the correct value for that measurement.

Precision of a Measurement

The **precision** of a measurement system refers to how close the agreement is between repeated measurements (which are repeated under the same conditions).

Accuracy vs. Precision

Accuracy

How closely a measurement agrees with the **true (accepted) value**.

Precision

How closely **repeated measurements** agree with each other.

A balance reads 1.001 g for a 1.000 g standard mass. $\Rightarrow$ **accurate**.

Same balance reads 0.900 g, 0.901 g, 0.900 g repeatedly. $\Rightarrow$ **precise but not accurate**.

High precision without accuracy = **systematic error** (instrument is consistently wrong).
Low precision = **random error** (irreducible scatter in measurements).

Check Your Understanding

An archer shoots five arrows at a target. All five cluster tightly together, but in the upper-left corner — not the bullseye.

Are the shots **accurate**, **precise**, both, or neither?

Come to lecture to see the answer.

Certainty in Measurements

Some measurements convey more confidence in their accuracy than others.

Which do you expect to be closer to 5 kilograms?:

- 5-kg Bag of Potatoes
- 5-kg Weight Plate used in an Olympic Weightlifting Competition

Significant Figures

We communicate this certainty by using significant figures.

- Mass of Potato Bag: 5 kg
- Mass of Weight Plate: 5.00 kg $\longleftarrow$ these extra zeroes are *significant*

Significant figures communicate how much certainty there is in measured values.

Significant Figures

1. **Rules for Counting Significant Figures** ***(sf or sig. figs)***
   - All non-zero digits are significant.
     (e.g. 123 has 3 sf)
   - Zeros between non-zero digits are significant.
     (e.g. 12304 has 5 sf)
   - Leading zeros are not significant.
     (e.g. 0.0012 has 2 sf)
   - Trailing zeros are significant only if a decimal point is present.
     (e.g. 120 has 2 sf; 120. has 3 sf; 120.30 has 5 sf)

Check Your Understanding

Determine the number of significant figures in the following measurements:

a. 0.0009
b. $15,450.0$
c. 6×10^3
d. 87.990
e. 30.42

Raise your hand when you have the answer.

Check Your Understanding

How many significant figures are in each measurement?

(a) 5.28 mL

(b) 5.28000 mL

(c) 0.04275

(d) 350. g (note the decimal point)

(e) 350 g (no decimal point)

Come to lecture to see the answer.

Significant Figures

2. Addition & Subtraction

- Round the result to the same decimal place as the **least precise** term.

  Keep the minimum number of digits after the decimal place in your answer.

$$12.11 + 0.3 + 1.025 = 13.435 \xrightarrow{\text{round to tenths}} 13.4$$

Sig. Figs. in Addition & Subtraction

Round the result to the same **decimal place** as the least precise term (fewest decimal places).

Examples

$$5.2 + 7.35 = 12.55 \xrightarrow{\text{round to tenths}} \boxed{12.6}$$

$$1.0023\text{ g} + 4.383\text{ g} = 5.3853\text{ g} \xrightarrow{\text{round to thousandths}} \boxed{5.385\text{ g}}$$

$$486\text{ g} - 421.23\text{ g} = 64.77\text{ g} \xrightarrow{\text{round to ones}} \boxed{65\text{ g}}$$

Addition/subtraction: consider **decimal places**, not total significant figures.

Check Your Understanding

Anna has two bags of oranges and one bag of apples.

Each bag of oranges weighs 13.52 lb, while the bag of apples weighs 10.2 lb.

What is the total weight of the bags?

Raise your hand when you have the answer.

Significant Figures

3. Multiplication & Division

- Round the result to the same *total significant figures* as the factor with the fewest.

$$4.56 \times 1.4 = 6.384 \xrightarrow{\text{round to 2 sf}} 6.4$$

Sig. Figs. in Multiplication & Division

Round the result to the same **number of significant figures** as the factor with the **fewest** significant figures.

Examples

$$5.0\ \frac{\mathrm{N}}{\mathrm{m}^2} \times 2.2\ \mathrm{m}^2 = 11.0\ \mathrm{N} \xrightarrow{\text{2 sig. figs.}} \boxed{11\ \mathrm{N}}$$

$$5\ \frac{\mathrm{N}}{\mathrm{m}^2} \times 2.2\ \mathrm{m}^2 = 11.0\ \mathrm{N} \xrightarrow{\text{1 sig. fig.}} \boxed{10\ \mathrm{N}}$$

5.0 has 2 sig. figs.; 5 has 1. The *least* precise factor limits the answer.

Check Your Understanding

A piece of aluminum foil measures $6.6\,\text{cm} \times 0.6238\,\text{cm}$.

What is its area, expressed to the correct number of significant figures?

Come to lecture to see the answer.

Example: Density with Significant Figures

Water Displacement Method — (chem2e_ch1_ex1.7)

A steel rebar has mass 69.658 g. It is placed in a graduated cylinder. The water level rises from 13.5 mL to 22.4 mL. Find the density of the rebar.

Volume displaced (subtraction):

$$V = 22.4 - 13.5 = 8.9\,\mathrm{mL}$$

(round to tenths; 2 sig. figs. in result)

$$d = \frac{m}{V} = \frac{69.658\,\mathrm{g}}{8.9\,\mathrm{mL}}$$

$$\boxed{d \approx 7.8\,\frac{\mathrm{g}}{\mathrm{mL}}}$$

Iron: $7.9\,\frac{\mathrm{g}}{\mathrm{mL}}$ $\therefore$ rebar is mostly iron.
(Limited to 2 sig. figs. by the volume measurement)

Check Your Understanding

Density ρ is equal to mass m divided by volume V.

$$\rho = \frac{m}{V}$$

A sample of an unknown liquid has a mass of $45.2\,\text{g}$ and occupies a volume of $38\,\text{mL}$. What is the density of the liquid? Express your answer in $\frac{\text{g}}{\text{mL}}$.

Raise your hand when you have the answer.

Example: Significant Figures

Considering Significant Figures in Calculations

What is the area of a circle 3.102 cm in diameter?

$$A_\circ = \pi r^2$$

$$A_\circ = \pi \left(\frac{d}{2}\right)^2$$

$$A_\circ = \pi \left(\frac{3.102\,\text{cm}}{2}\right)^2$$

$$A_\circ = 7.557\,418\,429\,1\,\text{cm}^2$$

$$A_\circ = 7.557\,\text{cm}^2$$

1. Count the significant figures (sf).
   there are 4 sf in 3.102 cm
2. Apply sf rules for multiplication.
   ∴ answer must have 4 sf.
3. Calculate answer to many digits.
4. Round to correct number of sf.

Percent Uncertainty of a Measurement

One method of expressing uncertainty is as a **percent of the measured value.**

If a measurement A is expressed with uncertainty δA, the percent uncertainty %unc is defined to be

$$\%\text{unc} = \frac{\delta A}{A} \times 100\%.$$

Example: Percent Uncertainty

Calculating Percent Uncertainty

A grocery store sells 5-lb bags of apples.
You purchase four bags over the course of a month and weigh the apples each time.
You obtain the following weight measurements per week: 4.8 lb, 5.3 lb, 4.9 lb, 5.4 lb.

You have previously determined, using additional measurements, that the uncertainty in the average weight of the 5-lb bag is ± 0.4 lb.

What is the **average weight** and **percent uncertainty** of the bag's weight, given these 4 measurements?

Average weight:

$$\frac{1}{4}(4.8 + 5.3 + 4.9 + 5.4)\,\text{lb} = \boxed{5.1\,\text{lb}}$$

$$\%\text{unc} = \frac{\delta A}{A} \times 100\%$$

$$\%\text{unc} = \frac{0.4\,\text{lb}}{5.1\,\text{lb}} \times 100\% = \boxed{8\%}$$

Example: Percent Uncertainty

$\therefore$ we can express the weight of a 5-lb bag of apples in terms of:

absolute certainty: $5.1\,\text{lb} \pm 0.4\,\text{lb}$

OR

percent uncertainty: $5.1\,\text{lb} \pm 8\%$

Check Your Understanding

A high-school track team's top sprinter clocked a 100-meter sprint at $12.04\,\mathrm{s}$ last week and at $11.96\,\mathrm{s}$ this week.

The manual of the stopwatch used states that the stopwatch has an uncertainty of $\pm 0.05\,\mathrm{s}$.

Can we conclude that the sprinter improved their time?

Raise your hand when you have the answer.

Percent Error in a Measurement

- Percent error measures how close an experimentally obtained value is to an accepted (theoretical) value.
- This shows the accuracy of a measurement.

$$\text{Percent Error} = \left| \frac{\text{Experimental Value} - \text{Accepted Value}}{\text{Accepted Value}} \right| \times 100\%$$

Or using a more compact form,

$$\%\ \text{Error} = \left| \frac{E - A}{A} \right| \times 100\%$$

Example: Percent Error

Calculating Percent Error in a Mass Measurement

Jessica measures the mass of a steel ball bearing using a balance. The accepted mass of the object is 22.4 g. She records a measurement of 21.28 g. What is the percent error of her measurement? Report your answer to two significant figures.

$$
\begin{aligned}
\%\ \text{Error} &= \left|\frac{E - A}{A}\right| \times 100\% \\
&= \left|\frac{21.28\ \text{g} - 22.4\ \text{g}}{22.4\ \text{g}}\right| \times 100\% \\
&= \left|\frac{-1.12\ \text{g}}{22.4\ \text{g}}\right| \times 100\% \\
&= 5.0\%
\end{aligned}
$$

The percent error of Jessica's measurement is **5.0%**. The negative sign is removed because percent error is always reported as an absolute value.

- A smaller percent error indicates a more accurate measurement.
- Units cancel in the calculation, so percent error is unitless.

Part IV

Mathematical Treatment of Measurement Results

Outline of Part IV: Mathematical Treatment of Measurement Results

Mathematical Treatment of Measurement Results

Outline of Part IV: Mathematical Treatment of Measurement Results

Dimensional Analysis

Strategy

- Write the given quantity with its unit.
- Multiply by a fraction where the unwanted unit is in the **denominator** and the desired unit is in the **numerator**.
- Units cancel like algebraic factors.
- Chain multiple conversion factors for multi-step problems.

Always check that your answer's unit makes sense. If it doesn't, the conversion factor was likely inverted.

Example: Mass Conversion

Converting Mass — (chem2e_ch1_ex1.8)

A frisbee has a mass of 125 g. What is its mass in ounces?
(Use: $1\,\text{oz} = 28.349\,\text{g}$)

$$125\,\text{g}\left(\frac{1\,\text{oz}}{28.349\,\text{g}}\right) = 4.41\,\text{oz}$$

$$\boxed{125\,\text{g} = 4.41\,\text{oz}}$$

- Grams cancel ($\text{g}/\text{g} = 1$).
- Result has 3 sig. figs. ($\because$ 125 has 3).

Example: Multi-Step Conversion

Fuel Economy — (chem2e_ch1_ex1.10)

A car travels 1250 km on 213 L of gasoline.
What is its fuel economy in miles per gallon?
(Use: $1\,\text{km} = 0.62137$ mi; $1\,\text{L} = 1.0567$ qt; $4\,\text{qt} = 1\,\text{gal}$)

$$d = 1250\,\text{km}\left(\frac{0.62137\text{ mi}}{1\,\text{km}}\right) = 777\text{ mi}$$

$$V = 213\,\text{L}\left(\frac{1.0567\,\text{qt}}{1\,\text{L}}\right)\left(\frac{1\,\text{gal}}{4\,\text{qt}}\right) = 56.3\text{ gal}$$

$$\text{mpg} = \frac{777\text{ mi}}{56.3\text{ gal}}$$

$$\boxed{\text{mpg} = 13.8\text{ miles/gallon}}$$

Chain multiple conversion factors together; verify that all intermediate units cancel correctly.

Check Your Understanding

Without calculating, which setup correctly converts 210 cm to meters?

(A) $210\,\text{cm} \times \dfrac{100\,\text{cm}}{1\,\text{m}}$

(B) $210\,\text{cm} \times \dfrac{1\,\text{m}}{100\,\text{cm}}$

Come to lecture to see the answer.

Check Your Understanding

A blood sample has a volume of $5.0\,\mu L$.

Set up (but do not solve) the conversion to express this volume in **liters**.

Come to lecture to see the answer.

Summary: Chapter 1

- **Chemistry**: the study of the composition, properties, and interactions of matter.
- Matter is classified as **pure substances** (elements, compounds) or **mixtures** (homogeneous, heterogeneous).
- Properties and changes are **physical** or **chemical**; properties are **extensive** or **intensive**.
- **SI units** and metric prefixes provide a universal system; density $d = m/V$ is intensive.
- **Temperature** conversions link Celsius, Fahrenheit, and Kelvin.
- **Significant figures** communicate the precision of measurements; apply different rules for addition/subtraction vs. multiplication/division.
- **Dimensional analysis** uses conversion factors to convert between units; unwanted units cancel.